

Armin Rahmati

Software Engineer



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portfolio



Profile

Backend & Automation Engineer — 3 yrs (1 year corporate, 2 years freelance) in Node.js/Go/FastAPI, passionate about system design principles, orchestrate workflows with n8n, and data analytics. Currently pursuing MSc at Amirkabir University.

Experience

OTICAMP | Head of Backend & Developer(FastAPI)

Jan 2026

- Architected a Modular Monolith system for a Radio Network Monitoring System (RNMS)
- Engineered a high-performance real-time layer using FastAPI WebSockets and Redis Pub/Sub, enabling efficient broadcasting of radio signal updates and ALE calls to multiple concurrent dashboard clients with sub-100ms latency.
- Optimized time-series data storage by TimescaleDB, enabling instant retrieval of historical signal quality trends and reducing query execution time for analytical reports.
- Standardized API documentation using Scalar OpenAPI, streamlining frontend integration and reducing developer onboarding time.
- establishing CI/CD workflows and conducting code reviews to enforce SOLID and clean code principles reducing production bugs.

Tasu Ventures | BackendDeveloper(Node.js)

Apr 2024 - oct 2024

- Developed a **web scraper** for data collection, bypassing **client rate limiting** (fine-tuning req per second), ensuring consistent data retrieval despite client restrictions.
- Developed an LLM-powered **AI chatbot** using RAG, indexing scraped data into PostgreSQL and enforcing business rules via strict prompt engineering, reducing user query resolution time.
- Managed daily **async tasks** via **BullMQ Redis queues**, maintaining <0.1% failure rates, improving system responsiveness. Integrated Instagram/Facebook APIs using **webhooks** with **challenge-response** auth, reducing API costs.
- Designed **Prometheus logging** with log-level filtering, **rotation** (50MB/30d retention), and conditional logging, reducing **storage costs**.
- Implemented **RBAC** with **middleware** validation for 4 roles (Super Admin, Admin, Client, User) eliminating unauthorized access Participated in **code reviews**, enforcing SOLID principles and reducing **tech debt**.
- Achieved full **test coverage** for components (unit/E2E).

Technology Development Center of Amirkabir University | Backend Developer (Node.js)

Jan 2023 - Apr 2023

Internship under the supervision of Dr. HamidReza Shahriri

Developed a **buyer's request management** system for locating nearby car spare parts.
Used PostgreSQL with **PostGIS** for efficient **spatial queries** (**Haversine formula**), enabling <50ms proximity searches.
Integrated **Neshan API** for optimized mapping and routing, reducing user traffic.
Reduced **notification system** memory footprint using **Redis pub/sub** with **message batching** and **TTL cleanup**.

Projects

Palette | AI Business Management Platform

AI-driven platform for online business management and social media engagement. It unifies inventory data from multiple sources for real-time insights, boosting efficiency, lowering support costs, and enhancing customer satisfaction.

Tech: Node.js, typescript, LangChain, BullMQ, Redis, ELK, Web Scraping (Puppeteer)

[visit](#)

Store Locator | Geospatial Parts Sourcing System

Implemented map-based store tracking and real-time inventory updates via Socket.io, reducing part search time

Tech: Node.js, Express, Docker, PostgreSQL/PostGIS, Redis, Neshan API, Socket.io

[github](#)

Birthday Celebration Automation | n8n

Developed and deployed an automated employee birthday recognition system which triggered personalized Telegram message to managers, streamlining the giftselectionanddistribution process.

Tech: N8N, Telegram, VPS Deployment, Automation

FundMe Blockchain Crowdfunding

Implemented smart contracts with Chainlink price feed integration for secure, real-time value conversion. Engineered backend services for contract deployment, funding, and withdrawal, and wrote comprehensive unit tests for all core features

Golang, Ethereum, smart contracts

[github](#)

Skills

- **Programming Languages**
 - Python, Go
- **Runtime Environments**
 - Node.js
- **Frameworks**
 - Express, Nest, FastAPI
- **Testing**
 - Functional (Unit & E2E)
 - Non-functional (Security, Performance)
- **Real-Time**
 - Socket.io
- **Web Scraping**
 - Puppeteer, Selenium
- **Version Control**
 - Git
 - Gitlab CI/CD Pipelines
 - Docker
- **Database Systems**
 - SQL: PostgreSQL, MySQL
 - NoSQL: Redis
 - Time-series: TimescaleDB

- **Monitoring & Observability**
 - Prometheus (Monitoring & Alerting)
 - Elasticsearch (Search Engine)
- **Background Jobs**
 - Celery

familiar with

- N8N (Workflow Orchestration)
- BPMN 2.0, Bizagi
- IOT(analyzing sensor data)
- Linux(server deployment, bash scripting, access control)
- Developing Telegram bots
- Data Analysis (Collection, Preprocessing, Visualization)
- Multithreading and Multiprocessing
- Cryptography Fundamentals
- System Architecture, Design Patterns, and Clean Code Practices
- NLP

Honors & awards

2nd Place – Codestar – Software Engineering Contest

Ranked #2 out of 938 competitors in a 3-hour, Solved 7 algorithms problems under strict time/memory limits using Python/C++.

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visit

Education

Amirkabir University of Technology (Tehran Polytechnic)	Babol Noshirvani University of Technology	September 2020
Masters degree, Computer cyber security September 2024	Bachelor's degree, Computer Engineering	September 2024